

Anjali Devi Sivakumar

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EDUCATION

University of Michigan, Ann Arbor, MI

Ph.D. in Electrical and Computer Engineering, GPA: 4.0 /4.0

April 2026 (Anticipated)

Dissertation: *Development of Miniaturized Gas Sensors for Biochemical and Biomedical Applications*

Advisors: *Prof. Xudong (Sherman) Fan and Prof. Zhaohui Zhong*

University of Michigan, Ann Arbor, MI

Master of Science in Electrical and Computer Engineering, GPA: 4.0 /4.0

April 2023

SASTRA Deemed to be University, Tamil Nadu, India

Bachelor of Technology in Electronics and Instrumentation Engineering, GPA: 9.31/10.0

July 2019

RESEARCH EXPERIENCE

University of Michigan, Dept. of Biomedical Eng. and Dept. of Electrical and Computer Eng.

Ann Arbor, MI

Graduate Student Research Assistant

Jan 2021- Present

Sensor Design and Characterization

- Spearheading collaboration with Southwest Research Institute (SwRI) to develop a robust MEMS-based μ GC suited for the Automated Lander Planetary Analog Composition Analyzer for extra-planetary exploration.
- Modelled and characterized a cost-effective, high-efficiency MEMS-based windowless helium dielectric barrier discharge μ PID that uses high-density needles to detect gases with an ionization potential up to 17.5 eV.
- Engineered and characterized low-temperature (<200 °C) Au-In-based TLP bonding methods for robust 3D hybrid integration of CTE-mismatched heterogeneous substrates.
- Designed and evaluated the performance MEMS-based differential dual-channel μ PID for high resolution (<10 ms) and sub-ppt VOC peak detection suitable for advanced μ GC systems.

In-situ Sensor/System Testing

- Led hardware and software development of a wearable rapid skin health tracker, successfully tested on 25+ subjects, for early detection of food anaphylaxis in collaboration with the Mary H. Weiser Food Allergy Center.
- Engineered a Wearable Analytical Skin Probe (WASP) resistant to environmental changes and a signal-processing algorithm to monitor skin emissions, tested on 150+ outpatients at Michigan Medicine to aid 20+ disease screenings.
- Assisted in the development of portable 1D μ GC exhaled breath analyzers and contributed to their adaptation for in-clinic use for COVID-19 detection, in collaboration with the Weil Institute, achieving a detection accuracy of ~94.7% within a cohort of 167 subjects.

Harvard Medical School – Brigham and Women’s Hospital, Division of Eng. in Medicine

Cambridge, MA

Research Assistant

Jan 2019 – Nov 2020

- Designed and prototyped a saliva- and smartphone-based, electronics-free, low-cost (<\$1) ovulation tracker for at-home use.
- Developed a fully integrated microfluidic point-of-care diagnostic platform for rapid and sensitive multiplex detection of SARS-CoV-2, HIV, HCV, and HBV.
- Spearheaded the development and calibration of smartphone-based semen analyzers (~\$4.45) that were successfully tested remotely by 92 subjects for the Reproductive Effects of Chemicals and Air Pollution (RECAP) study conducted by Harvard School of Public Health, achieving ~86.2% specificity for detecting low sperm concentration.
- Conducted histological analysis of 100+ semen and malaria blood cell samples using low-cost portable imaging systems to train adaptive adversarial neural network models for accurate morphological classification at low-image resolutions.

Indian Institute of Space Science and Technology, Dept. of Avionics

Trivandrum, India

Summer Research Fellow

May 2018 – July 2018

- Developed and evaluated a cost-effective two-stage amplifier based-analog readout for low-valued resistive Pd-based H_2 gas sensors, demonstrating operational robustness with a linear output and zero offset.
- Designed and characterized a modified dual-slope ADC based on direct-digital front ends, low-valued resistive Pd-based H_2 gas sensors, showcasing the capability to generate a linear output unaffected by switch ON-resistances.

LEADERSHIP AND MENTORING EXPERIENCE

Graduate Society of Women Engineers (GradSWE), University of Michigan

Ann Arbor, MI

Senior K12 Outreach Officer

May 2023- Present

- Organized and led robotics workshop modules for Junior Girl Scouts, engaging over 20 participants during Winter 2024, Fall 2024, and Winter 2025, sponsored by Aptiv and the UM Center for Educational Outreach (CEO).
- Co-organized Explorer Day sponsored by Aptiv in Fall 2024, welcoming children from the Community Action Network (CAN), which supports under-resourced families in Washtenaw County; the event featured a visit to the UM campus and the Natural History Museum.
- Collaborated with Tau Beta Pi and the Mechanical Engineering Graduate Student Council to orchestrate Engineering Day for Brownie and Junior Girl Scouts, providing hands-on activities to introduce various engineering disciplines.

University of Michigan Undergraduate Research Opportunity Program (UROP)

Ann Arbor, MI

Research Mentor

Sept 2023 – April 2025

- Mentored four freshmen in engineering, guiding them in developing and characterizing portable gas analyzers for medical diagnostics.
- Advised undergraduate students on conducting research and presenting findings at symposium.

SKILLS

Microfabrication: CVD, PVD, ALD, Wafer/piece Bonding (Anodic/Eutectic/TLP), Dry and Wet Etching, Lithography, Dicing, Flip-Chip Bonder

Metrology: SEM/EDS, Stylus profilometry, Optical 3D profilometry, Optical Microscopy, Reflectometry, Ellipsometry, Goniometer, Four-point probe

Design Software: AutoCAD Fusion, KiCAD, Altium, Klayout

Programming Languages: C, C++, Python, Embedded C, MATLAB, Simulink, LabVIEW

Simulation Software: LT-Spice, TI-Tina, Cadence Virtuoso, COMSOL

Rapid Prototyping Hardware: CO₂ Laser cutting, 3-D Printing (SLA & FDM), CNC Machining, Water-jet cutter

Wet Lab: Histology, Cell culture, Biosafety Level 2+

JOURNAL PUBLICATIONS

- Anjali Devi Sivakumar**, Ruchi Sharma, Junqi Wang, Weiliang Liang, Yulong Zhou, and Xudong Fan, “Dual-Channel Microfluidic Photoionization Detector”, *Analytical Chemistry* **97**, 22397-22406, October 2025.
DOI: <https://doi.org/10.1021/acs.analchem.5c05021>
- Makiko Mitsunami, Hemanth Kandula, **Anjali Devi Sivakumar**, Tejas Girish Deshpande, Jasmine Kumar, Susitra Gnanasambhandam, Prudhvi Thirumalaraju, Manoj Kumar Kanakasabapathy, Charles L Bormann, Martin Kathrins, Hadi Shafiee, Jorge E Chavarro, Jaime E Hart, “Comparing remote smartphone-based semen assessments with laboratory evaluations in men unselected for fertility concerns”, *Fertility and Sterility*, Volume 124, Issue 6, 1245-1254, December 2025.
DOI: <https://doi.org/10.1016/j.fertnstert.2025.07.002>
- Xiaheng Huang, Shuo Yang, Wencheng Li, Robert Nidetz, Ruchi Sharma, **Anjali Devi Sivakumar**, Chandrakalavathi Thota, Hongbo Zhu, Weishu Wu, Seong-Yong Jeong, and Xudong Fan, “Microfluidic integration of μ PID on μ column for ultracompact micro-gas chromatography”, *Sensors and Actuators B: Chemical* **410**, 135717, July 2024.
DOI: <https://doi.org/10.1016/j.snb.2024.135717>
- Anjali Devi Sivakumar**, Ruchi Sharma, Chandrakalavathi Thota, Ding Ding, and Xudong Fan, “WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss”, *ACS Sensors* **8**, 4407-4416, November 2023.
DOI: <https://doi.org/10.1021/acssensors.3c01936>
- Xiaheng Huang, Ruchi Sharma, **Anjali Devi Sivakumar**, Shuo Yang, and Xudong Fan, “Ultrathin silica integration for enhancing reliability of microfluidic photoionization detectors”, *Analytical Chemistry* **95**, 8496–8504, May 2023.
DOI: <https://doi.org/10.1021/acs.analchem.3c00354>
- Ruchi Sharma, Wenzhe Zang, Ali Tabartehfarahani, Andres Lam, Xiaheng Huang, **Anjali Devi Sivakumar**, Chandrakalavathi Thota, Shuo Yang, Robert P. Dickson, Michael W. Sjoding, Erin Bisco, Carmen Colmenero Mahmood, Kristen Machado Diaz, Nicholas Sautter, Sardar Ansari, Kevin R. Ward, and Xudong Fan, “Portable Breath-Based Volatile Organic Compound Monitoring for the Detection of COVID-19 During the Circulation of the SARS-CoV-2 Delta Variant and the Transition to the SARS-CoV-2 Omicron Variant”, *JAMA Network Open* **6**, e230982, February 2023.
DOI: <https://doi.org/10.1001/jamanetworkopen.2023.0982>

7. Aridany Suárez-Trujillo, Hemanth Kandula, Jasmine Kumar, **Anjali Devi**, Larissa Shirley, Prudhvi Thirumalaraju, Manoj Kumar Kanakasabapathy, Hadi Shafiee, Liane Hart, “Validation of a smartphone-based device to measure concentration, motility, and morphology in swine ejaculates”, *Translational Animal Science* 6, 1-8, October 2022.
DOI: <https://doi.org/10.1093/tas/txac119>
8. Xiaheng Huang, Maxwell Wei-hao Li, Wenzhe Zang, Xiaolu Huang, **Anjali Devi Sivakumar**, Ruchi Sharma, and Xudong Fan, “Portable Comprehensive Two Dimensional Micro-Gas Chromatography Using Integrated Flow-Restricted Pneumatic Modulator”, *Nature Microsystems & Nanoengineering* 8, 115, November 2022.
DOI: <https://doi.org/10.1038/s41378-022-00452-5>
9. Manoj Kumar Kanakasabapathy, Prudhvi Thirumalaraju, Hemanth Kandula, Fenil Doshi, **Anjali Devi Sivakumar**, Deeksha Kartik, Raghav Gupta, Rohan Pooniwalla, John A Branda, Athe M Tsibris, Daniel R. Kuritzkes, John C Petrozza, Charles L Bormann, Hadi Shafiee, “Adaptive Adversarial Neural Networks for Lossy and Domain-Shifted Medical Image Analysis”, *Nature Biomedical Engineering* 5, 571-585, June 2021.
DOI: <https://doi.org/10.1038/s41551-021-00733-w>

CONFERENCE PROCEEDINGS AND PRESENTATIONS

1. **Anjali Devi Sivakumar**, Ruchi Sharma, Chandrakalavathi Thota, Anjali Sundar, Charles Schuler, and Xudong Fan, “WASP: A Next-Generation Wearable Solution for Personalized Skin Health Monitoring”, *Pittcon 2026*, San Antonio, USA, March 2026. (Accepted)
2. **Anjali Devi Sivakumar**, Ruchi Sharma, and Xudong Fan, “High Performance Microfluidic Dual Channel Photoionization Detector for Trace Vapor Detection”, *Pittcon 2026*, San Antonio, USA, March 2026. (Accepted)
3. Anjali Sundar, **Anjali Devi Sivakumar**, Lea Franco, Bridgette Kaul, Xudong Fan, Charles F. Schuler, “Utilizing a Novel Wearable Water Loss Monitor to Detect and Predict Food Anaphylaxis”, *American Academy of Allergy, Asthma & Immunology Annual Meeting*, Philadelphia, USA, February 2026. (Accepted).
4. **Anjali Devi Sivakumar**, Ruchi Sharma, Chandrakalavathi Thota, Shuo Yang, Ali Tabartehfarahani, Xudong Fan, “WASP: A Next-Generation Wearable Solution for Dynamic Skin Health Monitoring”, *UM Skin Research Center's 6th Annual Symposium: Advances in Skin Biology & Diseases*, Ann Arbor, USA, February 2025.
5. **Anjali Devi Sivakumar**, Ruchi Sharma, Chandrakalavathi Thota, Xudong Fan, “WASP: Wearable Analytical Skin Probe”, *46th Annual International Conference of the IEEE Engineering in Medicine & Biology Society*, Orlando, USA, July 2024.
6. Ruchi Sharma, **Anjali Devi Sivakumar**, Chandrakalavathi Thota, Ali Tabartehfarahani, Shuo Yang, Loc Cao, Brittany Baur, Haolin Li, Sardar Ansari, Zhaohui Zhong, and Xudong Fan, “Wearable Micro-GC Device for Non-Invasive Monitoring of Sweat VOC Patterns: Advancements in Disease Diagnosis”, *Breath Summit 2024: International Congress For Breath Research (IABR)*, Indiana, USA, June 2024.
7. Haolin Li, Ruchi Sharma, Xiaheng Huang, Chandrakalavathi Thota, **Anjali Devi Sivakumar**, Xudong Fan, Zhaohui Zhong, “Programmable Graphene e-Nose Sensor Arrays for Fast, Sensitive, and Label-free Identification of Chemical Vapors”, *MRS Spring Meeting*, Seattle, USA, April 2024.
8. Haolin Li, Ruchi Sharma, Xiaheng Huang, Chandrakalavathi Thota, **Anjali Devi Sivakumar**, Xudong Fan, Zhaohui Zhong, “Programmable Graphene e-Nose Sensor Arrays for Fast, Sensitive, and Label-free Identification of Chemical Vapors”, *245th ECS Meeting*, San Francisco, USA, May 2024.
9. Xiaheng Huang, Shuo Yang, Wencheng Li, Robert Nidetz, Ruchi Sharma, **Anjali Devi Sivakumar**, Chandrakalavathi Thota, Hongbo Zhu, Weishu Wu, Seong-Yong Jeong, Xudong Fan, “Monolithically Integrated μ PID on μ COLUMNS for Ultracompact Micro-Gas Chromatography”, *37th International Conference on Micro Electro Mechanical Systems (MEMS)*, Austin, USA, January 2024.
10. Loc Cao, Wenzhe Zang, Ruchi Sharma, Ali Tabartehfarahani, Chandrakalavathi Thota, **Anjali Devi Sivakumar**, Andres Lam, Xudong Fan, Kevin R Ward, Sardar Ansari, “Automated Gas Chromatography Peak Alignment: A Deep Learning Approach using Greedy Optimization and Simulation”, *45th Annual International Conference of the IEEE Engineering in Medicine & Biology Society*, Sydney, Australia, July 2023.
11. Ahmed Shokr, Luis G. C. Pacheco, Prudhvi Thirumalaraju, Manoj K. Kanakasabapathy, **Anjali Devi Sivakumar**, Jahnvi Gandhi, Deeksha Kartik, Filipe S. R. Silva, Eda Erdogmus, Sai Hemanth K. Kandula, Shenglin Luo, Xu G. Yu, Raymond T. Chung, Jonathan Z. Li, Daniel R. Kuritzkes, and Hadi Shafiee, “Rapid, Sensitive, Point-of-Care Detection of SARS-CoV-2 On-Chip”, *Discover Brigham*, Virtual, November 2020.
12. **Anjali Devi Sivakumar** and Anoop C.S., “Analog and Direct-Digital Front-End Circuits Suitable for Low-Valued Resistive Sensors”, *15th IEEE India Council International Conference*, Coimbatore, India, December 2018.

PATENTS

1. Xudong Fan, **Anjali Devi Sivakumar**, “Dual-channel photoionization detector and methods for use thereof”, (Provisional patent application no. 63/890,891, filed on September 30, 2025).
2. Xudong Fan, **Anjali Devi Sivakumar**, “Modifications to TEWL device”, 2025 (Patent disclosure filed).
3. Xudong Fan, Ruchi Sharma, Chandrakalavathi Thota, **Anjali Devi Sivakumar**, Ali Tabartehfarahani, Shuo Yang, Wencheng Li, Loc Cao, Sardar Ansari, “Skin emission sampling pouch”, 2024 (Patent disclosure filed).
4. Xudong Fan, Xiaheng Huang, Shuo Yang, Wencheng Li, Robert Nidetz, Ruchi Sharma, **Anjali Devi Sivakumar**, Chandrakalavathi Thota, Hongbo Zhu, Weishu Wu, Seong-Yong Jeong, “Monolithic integration of micro-gas chromatography column and micro-photoionization detector on a chip”, (Provisional patent application number: 63/622,741, filed on January 19, 2024).
5. Xudong Fan, **Anjali Devi Sivakumar**, “A transepidermal water loss device” (PCT application number: 18/428,514, filed on February 1, 2024).
6. Xiaheng Huang, Ruchi Sharma, **Anjali Devi Sivakumar**, Shuo Yang, and Xudong Fan, “Protective Dielectric Layers for Enhancing Reliability of Vacuum Ultraviolet Lamps” (PCT application number: PCT/US2024/022212, filed on March 29, 2024).

ACADEMIC HONORS AND AWARDS

- Received the **Rackham International Student Fellowship** of \$14,596 for Winter 2026.
- Secured the **Rackham Graduate Student Research Grant** of \$3,000 to support my research on Au-In TLP bonding.
- Honored with the **Dr. Helen Wu Award** of \$8,000 from Rackham Graduate School for Spring/Summer 2024.
- Awarded the distinguished **Barbour Scholarship** (established in 1917) from Rackham Graduate School, covering full tuition and a stipend of \$36,084 for the 2024-25 academic year.
- Received the **Rackham Conference Travel Grant** of \$900 for the 2024-25 and 2025-26 academic years.
- Inducted into the **Excellence in ECE Honor Roll 2023-24** for contributions to departmental events and initiatives fostering an inclusive community.
- Awarded the **Research Excellence Award** from **Brigham Research Institute** for **outstanding contributions to research** at **Brigham and Women’s Hospital** on Nov-12,2020.
- Awarded the **Desh Videsh Scholarship** for undergraduate thesis research at Harvard Medical School, MA.
- Honored with the **Smt. K. Thulasi Memorial Award for best outgoing student** in the 2019 graduating class of the School of Electrical and Electronics Engineering.
- Received the **Dr. V. Jayashankar Award** for highest CGPA (up to VII Semester) in the B.Tech.-Electronics and Instrumentation Engineering & M.Tech (five-year integrated) Instrumentation & Control program.
- Awarded the **Shri Vishwantha Iyer Award for best outgoing student** from the Department of Electronics and Instrumentation Engineering among the 2019 graduating batch.
- Awarded the **Shri. A.Venkataraman Award** for academic excellence from SASTRA Deemed to be University.
- Recognized with the **Dean’s Merit Scholarship** for ranking in the top 2% of the undergraduate class (2015-2018) based on academic performance.

PROFESSIONAL AND SERVICE ACTIVITIES

Nature Microsystems & Nanoengineering, *Reviewer*

University of Michigan UROP Symposium, *Judge*

Tau Beta Pi (TBP) - The Engineering Honor Society, *Active Member*

Eta Kappa Nu (HKN) – The Electrical and Computer Engineering Honor Society, *Active Member*

Spring 2024

Winter 2022 – Present

Fall 2021- Present